

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	Using Electroencephalography and Inverse Reinforcement Learning in order to develop and validate a multimodal platform for impulsivity research
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Chosen major:	Artificial Intelligence
Supervisor	Tomas Ward
Date of Submission	24/11/2025

Section 2: About your Practicum

Please answer all questions below. Please pay special attention to the word counts in all cases.

What is the topic of your proposed practicum? (100 words)

This practicum will develop a high-quality software platform that integrates Electroencephalography (EEG) with behavioural data captured during a serious-game driving task. Using Inverse Reinforcement Learning (IRL), the platform will extract impulsivity-related features from participants' decision-making patterns. While not intended for diagnosing ADHD or any clinical condition, the system is designed as a robust foundation for future data-collection studies involving attentional or impulsivity-related disorders. Initial evaluation will use non-clinical participants, leveraging natural variability in impulsivity. Their EEG and behavioural metrics will be compared with validated self-report measures such as the Barratt Impulsiveness Scale (BIS-11) to assess meaningful correlations.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. **Bridging the Gap Between Self-Report and Behavioral Laboratory Measures: A Real-Time Driving Task with Inverse Reinforcement Learning** (by Sang Ho Lee, Myeong Seop Song, Min-hwan Oh, and Woo-Young Ahn):

- a. In this paper, the user engages with a Highway task, in which they would dodge cars and make decisions (by changing lanes for example) in order to drive safely. If the user crashed too many times, and their fuel ran out, then the task would end.
2. **Modeling sensory-motor decisions in natural behavior** (by Ruohan Zhang, Shun Zhang, Matthew H. Tong, Yuchen Cui, Constantin A. Rothkopf, Dana H. Ballard, Mary M. Hayhoe)
 - a. In this paper, they aim to prove that an inverse reinforcement model is appropriate over other machine learning methods, and can be developed in order to predict the rewards from human behavioural data.
3. **fNIRS-EEG BCIs for Motor Rehabilitation: A Review** (by Jianan Chen, Yunjia Xia, Xinkai Zhou, Ernesto Vidal Rosas, Alexander Thomas, Rui Loureiro, Robert J. Cooper, Tom Carlson and Hubin Zhao)
 - a. This paper details the usefulness of Electroencephalograms and Brain Computer Interfaces for motor rehabilitation by recording brain signals and using that data to develop Brain Computer Interface applications that control some application.
4. **Virtual Reality for Enhanced Ecological Validity and Experimental Control in the Clinical, Affective and Social Neurosciences** (by Thomas D. Parsons)
 - a. This is about Using Virtual Reality to simulate real world conditions. This paper in particular mentions one researcher, who dismisses the claims of another researcher, on the basis that their experiment isn't realistic or representative of the real world.
5. **Neurophysiological methods for monitoring brain activity in serious games and virtual environments: a review** (by Manuel Ninaus, Silvia Erika Kober, Elisabeth V.C. Friedrich, Ian Dunwell, Sara de Freitas, Sylvester Arnab, Michela Ott, Milos Kravcik, Theodore Lim, Sandy Louchart, Francesco Bellotti, Anna Hannemann, Alasdair G. Thin, Riccardo Berta, Guilherme Wood, Christa Neuper)
 - a. A review of neuroscientific methods and techniques used in conjunction with serious games / game-based learning.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

Our work extends ideas from the first paper, which used IRL to analyse driving-task decisions, by integrating EEG recordings to create a richer multimodal dataset. Whereas previous studies aimed to estimate behavioural traits directly, our focus is the creation of a high-fidelity research platform capable of capturing synchronised EEG and behavioural data in an ecologically valid environment. The second paper demonstrates how IRL can uncover latent reward structures from observable behaviour, informing our modelling approach for impulsivity-related actions. Insights from the third paper guide our EEG acquisition pipeline, including considerations of signal quality, artefact management, and data synchronisation. The fourth paper highlights the value of realistic virtual environments for cognitive research, supporting our use of a visually immersive serious-game driving task. The final paper offers an overview of how neurophysiology and serious games can be combined, aligning closely with our goal of producing a platform suited for future clinical and experimental research. Together, these works justify our methodological choices while clarifying that the contribution lies in software and data-collection infrastructure rather than clinical diagnosis.

What are the research questions that you will attempt to answer? (200 words)

This practicum does not attempt to diagnose ADHD. Instead, it focuses on developing and validating a multimodal platform for impulsivity research. The central research questions are:

1. **Can a combined EEG and behavioural data-collection platform reliably capture high-quality, synchronised information during a serious-game driving task?**
2. **Can IRL-derived features, such as inferred reward functions and decision policies, serve as meaningful indicators of impulsivity in non-clinical participants?**
3. **Do impulsivity metrics generated by the platform correlate with standardised self-report measures such as the Barratt Impulsiveness Scale (BIS-11)?**
4. **Does the platform demonstrate potential for future research on attentional and impulsivity-related conditions, even without access to clinical populations?**

These questions ensure the project remains feasible, ethical, and scientifically grounded while producing a system that can support future large-scale or clinical studies.

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- What software and programming environment will you use?
 - o Unreal Engine 5, Visual Studio (C++), and OpenVibe for Lab Streaming Layer (LSL) integration.
- What coding/development will you do?
 - o Development of the driving-task environment; implementation of IRL algorithms; construction of a pipeline for acquiring EEG signals, synchronising them with behavioural events, and generating impulsivity-related metrics.
- What data will be used for your investigations?
 - o Data will be collected from non-clinical participants performing the driving task while wearing an EEG device.
- Is this data currently available, if not, where will it come from?
 - o All data will be generated by the research team during platform testing.
- What experiments do you expect to run?
 - o Participants complete the driving task while EEG and behavioural metrics are recorded. Afterwards, they complete the Barratt Impulsiveness Scale (BIS-11). IRL-derived and EEG-derived features will be compared with self-reported impulsivity scores.
- What output do you expect to gather?
 - o A functioning multimodal platform; a synchronised pilot dataset; and correlation analyses between IRL/EEG features and BIS-11 impulsivity scores.
- How will the results be evaluated?

- Evaluation will focus on data quality, system reliability, and the strength of correlations between platform-derived metrics and BIS-11 scores. No clinical diagnostic accuracy will be assessed.